Advanced Databases : **Project Part 1 – Data Modeling and Database Design**

Title : **Airline Company Database Design**

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1. Summary

An airline company database is a complex information management system designed to organize and manage an enormous amount of data essential for an airline's efficient operation. This project explores the world of databases, focusing on critical entities such as airline companies, staff, aircrafts, airports, passengers, flights, tickets and their relationships.

The study explores one-to-many relationships, analyzing how an airline firm owns several planes, employs numerous individuals and schedules different flights. Many-to-many connections further emphasize the complexity, showing the system that connects airports, airline companies and passengers and their interactions with flights. It underlines how flexible and interconnected the aviation sector is.

In this paper we also carefully analyze data types, primary keys, foreign keys, as well as the entity-relashionship diagram and the relational model of this specific database.

2. Sample Queries

- Names of all passengers who have booked a flight for 25/12.
- IDs of passengers who live in Canada and travel to Spain.
- Find the IDs of airline employees who are over 50 years old.
- Find the name of the youngest employee who works as a pilot.
- What is the total number of staff that the airline company employs?
- Which are the airports that are located in Greece?
- Which tickets cost over 300 euros?
- What is the average ticket cost for every flight?
- How many flights will be scheduled during 25/12 – 31/12?
- Models of airplanes manufactured after 2010 and have capacity over 100 seats?

3. Data Requirements

3.1 Entities

- **AIRLINE COMPANY**: This entity represents the airline company and its main information.
- **STAFF**: This entity represents the employees who work for this airline company.
- **AIRPLANE**: This entity represents the aircraft that belongs to the airline company.
- **AIRPORT**: This entity represents the airport with which the airline company cooperates.
- **PASSENGER**: This entity shows us the passengers/customers who has used or are going to use this airline company.
- **FLIGHT**: This entity represents the flight that is about to happen.
- **TICKET**: This entity informs us about the tickets that passengers of this airline will use.

3.2 Attributes and Data Types

- **AIRLINE COMPANY**

  cid (integer): the id number of the airline company
  cname (varchar): the name of the airline company
  cphone (integer): the phone number of the airline company
  cemail (varchar): the email address of the airline company

- **STAFF**

  sssn (integer): the social security number of the employee of the airline company
  sname (varchar): the name of the employee
  sphone (integer): the phone number of the employee
  sjob title (varchar): the job title of the employee
  sage (integer): the age of the employee

- **AIRPLANE**

  pid (integer): the id number of the airplane that belongs to the the airline company
  pmodel (varchar): the model type of the airplane
  pyear (integer): the year that the airplane was manufactured
  pcapacity (integer): the capacity number of the airplane

- **AIRPORT**

  aid (integer): the id number of the airport an airline company connects with
  aname (varchar): the name of the airport
  alocation (varchar): the location of the airport

- **PASSENGER**

  pssn (integer): the social security number of the passenger that will fly with the airline company
  pname (varchar): the name of the passenger
  pphone (integer): the phone number of the passenger
  paddress (varchar): the address of the passenger
  age (integer): the age of the passenger

- **FLIGHT**

  fid (integer): the id number of the flight
  fdate (date): the date of the flight
  fdestination (varchar): the destination of the flight

- **TICKET**

  tid (integer): the id number of the ticket
  tseat (integer): the seat that the ticket books
  tcost (integer): the cost of the flight

3.3 Relationships

3.3.1 One-to-many

- **AIRLINE COMPANY - AIRPLANE**
  An airline company owns multiple airplanes and an airplane belongs to one airline company

- **AIRLINE COMPANY - STAFF**
  An airline company employees multiple individuals but an individual can only work for one airline company

- **AIRLINE COMPANY - FLIGHT**
  An airline company has multiple flights and a flight isabout one airline company

- **FLIGHT – TICKET**
  A flight has many tickets but a ticket corresponds to one flight

- **PASSENGER – TICKET**
  A passenger can book many tickets but a ticket corresponds to one passenger

3.3.2 Many-to-many

- **AIRLINE COMPANY - AIRPORT**
  An airline company connects with many airports and an airport has connections with many airline companies

- **PASSENGER – FLIGHT**
  Passengers may book different flights and flights can have many passengers

3.4 Constraints

3.4.1 Primary Keys

- **Airline Company – CID**: Uniquely identifies each airline company
- **Airplane - PID**: Uniquely identifies each airlplane
- **Staff - SSSN**: Uniquely identifies each staff member
- **Airport - AID**: Uniquely identifies each airport
- **Flight - FID**: Uniquely identifies each flight
- **Passenger - PSSN**: Uniquely identifies each passenger
- **Ticket - TID**: Uniquely identifies each ticket

3.4.2 Foreign Keys

- **Airplane - CID**: Has as foreign key the primary key of the airline company
- **Staff - CID**: Has as foreign key the primary key of the airline company
- **Flight - CID**: Has as foreign key the primary key of the airline company
- **Ticket – FID and PSSN**: Has as foreign keys the primary keys of the flight and the passenger

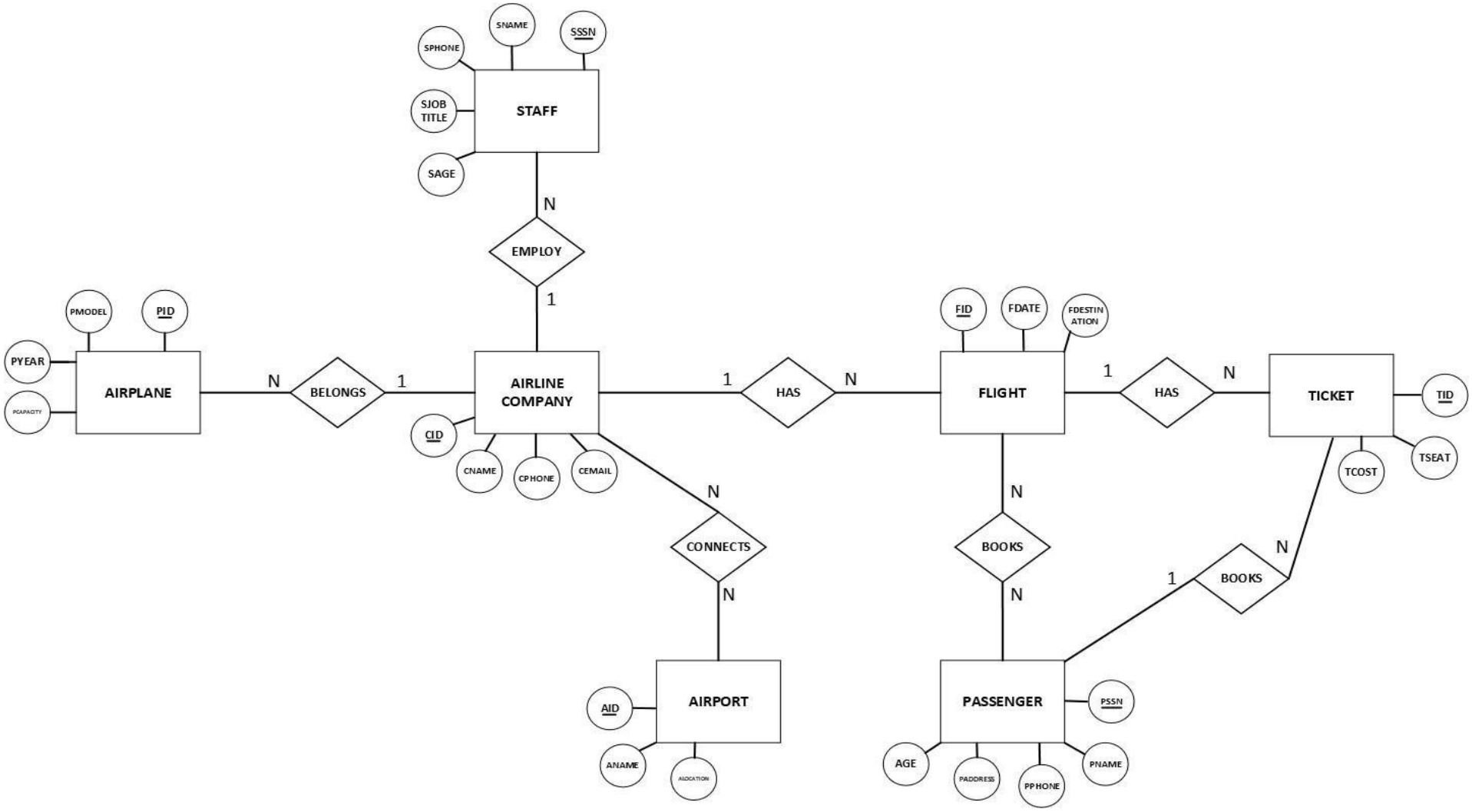
3.4.3 Not Null

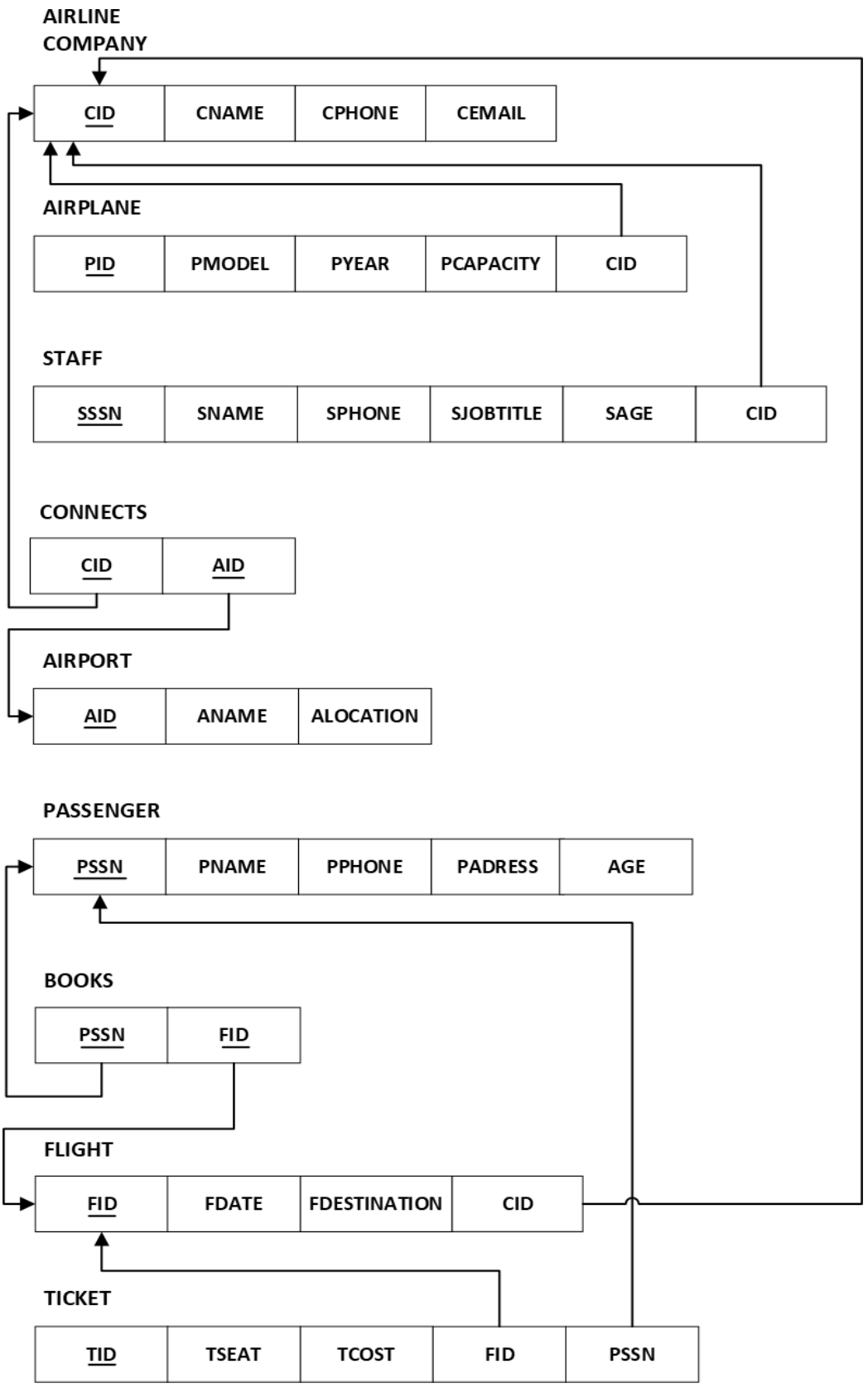
Primary keys must be not null

3.4.4 Unique

Primary keys are unique by default

4. ER-diagram



Note: The constraints (except the primary keys) and the data types can not be modelled by the ER-diagram

5. Relational Model